

AUTOMATED ATTENDANCE EXTRACTION AND REPORT GENERATION SYSTEM

**23CS1612-PROFESSIONAL READINESS FOR INNOVATION,
EMPLOYABILITY AND ENTREPRENEURSHIP**

A PROJECT REPORT

Submitted by

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IN

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CHENNAI- 600123.**

(An Autonomous Institution Affiliated to Anna University, Chennai)

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PANIMALAR ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

BONAFIDE CERTIFICATE

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Signature of the staff incharge with date

<<Name of the staff incharge >>

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professional readiness for innovation, employability and entrepreneurship examination held

on.....

INTERNAL EXAMINER

EXTERNAL EXAMINER

DECLARATION BY THE STUDENT

We **ARCHANA A S (211423104050)**, **APARNA P (211423104049)** hereby declare that this project report titled **“AUTOMATED ATTENDANCE EXTRACTION AND REPORT GENERATION SYSTEM”**, under the guidance of **“NAME OF THE PROJECT INCHARGE”** is the original work done by us and we have not plagiarized or submitted to any other degree in any university by us.

1. ARCHANA A S

2. APARNA P

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ABSTRACT

The Automated Attendance Extraction and Report Generation System is an automation-based solution developed to simplify and improve attendance management in educational institutions. Manual attendance tracking using registers or spreadsheets is time-consuming and prone to errors. This system automates the process of extracting attendance data from Excel or databases, validating the data, calculating attendance percentages, and generating structured reports. Using automation tools such as UiPath, the system ensures accurate calculations and eliminates repetitive manual tasks. It also identifies students below attendance thresholds and produces daily, weekly, and monthly reports for better monitoring. Logging and error-handling mechanisms ensure reliability. Overall, this system enhances efficiency, reduces human errors, and provides a scalable solution for managing attendance effectively.

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CHAPTER – 1

INTRODUCTION

CHAPTER – 1

INTRODUCTION

1.1 OVERVIEW OF PROJECT

Our project, "Automated Attendance Extraction and Report Generation System," is developed using UiPath Studio to enhance the efficiency and accuracy of attendance management in educational institutions. In many institutions, attendance is recorded and processed manually, which leads to errors in calculation, delays in report generation, and increased workload for faculty and administrative staff. This project focuses on automating key tasks such as extracting attendance data from Excel or databases, validating records, calculating attendance percentages, and generating structured reports.

The system is designed using a rule-based approach, where attendance criteria such as total working days, present days, and attendance thresholds are maintained in a structured format. This makes the solution flexible, easy to update, and scalable for real-world applications. By automating repetitive tasks, the system reduces human errors and ensures accurate and consistent attendance calculations for all students.

In addition to core functionalities, the project also includes features like data validation, error handling, and automated report generation. These reports provide better visibility into student attendance, helping faculty and management monitor performance and identify students with low attendance effectively. Overall, this project improves efficiency, reduces manual effort, and provides a reliable and scalable solution for managing attendance data in an accurate and organized manner.

1.2 PROBLEM DEFINITION

Current attendance management processes in educational institutions rely heavily on manual methods, including recording attendance, calculating percentages, and generating reports. This manual approach leads to inefficiencies, delays, and increased workload for faculty and administrative staff.

- Manual handling of attendance data often results in inconsistencies, as human errors may occur while entering data or calculating attendance percentages. This affects the accuracy and reliability of the overall system.
- The increasing number of students makes it difficult to manage attendance efficiently using manual methods. This results in delays in report generation, incorrect calculations, and reduced productivity.
- As institutional requirements grow, the absence of an automated system creates scalability challenges and limits the ability to track, analyze, and manage attendance data effectively.

CHAPTER 2

LITERATURE SURVEY

CHAPTER – 2

LITERATURE SURVEY

The rapid advancement of digital technologies has significantly transformed administrative processes across various domains, particularly in educational institutions. Attendance management systems, which traditionally relied on manual processes, are increasingly being automated using modern technologies such as Robotic Process Automation (RPA), data analytics, and intelligent workflow systems. A comprehensive literature review reveals that automation not only improves efficiency but also enhances accuracy, scalability, and transparency in attendance-related operations.

Brown and Williams (2019) [1] highlight the role of RPA in automating repetitive and rule-based tasks such as data extraction, validation, and report generation. Their study emphasizes that automation reduces human intervention, thereby minimizing errors and increasing workflow efficiency. Similarly, Patel and Gupta (2020) [2] discuss how automation significantly enhances the speed and accuracy of handling large datasets. This is particularly relevant in attendance systems, where large volumes of student data must be processed consistently and efficiently.

Garcia and Martinez (2018) [3] explore current trends in automation and stress the importance of scalability in system design. They argue that modern attendance systems must be capable of handling increasing data loads and user demands without performance degradation. Yang and Wang (2017) [4] further elaborate on advancements in RPA technologies, highlighting their ability to replicate human interactions with digital systems. This enables automated processes such as logging into portals, extracting attendance records, and updating databases with high consistency.

Kim and Park (2019) [5] focus specifically on the use of UiPath Studio for building automation workflows. Their research demonstrates how UiPath can effectively process structured data, making it suitable for attendance management tasks such as report generation and data validation. Chen and Li (2021) [6] address challenges associated with automation systems, including error handling, exception management, and system integration. They emphasize the importance of designing robust systems capable of handling unexpected scenarios to ensure reliability.

Singh and Sharma (2022) [7] present recent advancements in workflow automation within academic systems. Their study shows that automated attendance extraction and reporting significantly improve productivity, accuracy, and transparency. These findings directly support the implementation of automated attendance systems in educational institutions.

In addition to these foundational studies, Kumar and Reddy (2021) [8] examine the integration of automation with cloud-based platforms. They highlight that cloud integration enables real-time access to attendance data and improves system scalability. Similarly, Zhao and Chen (2020) [9] explore the role of data analytics in attendance systems, demonstrating how automated data processing can generate insights such as attendance trends and student performance patterns.

Almeida and Costa (2019) [10] discuss the importance of system interoperability, emphasizing that automated attendance systems should seamlessly integrate with existing Learning Management Systems (LMS) and Enterprise Resource Planning (ERP) systems. This integration ensures efficient data flow and reduces redundancy.

CHAPTER 3

SYSTEM ANALYSIS

CHAPTER – 3

SYSTEM ANALYSIS

3.1 EXISTING SYSTEM

Currently, attendance management in many educational institutions relies heavily on manual processes using registers or basic spreadsheet systems. Tasks such as recording daily attendance, verifying data, calculating attendance percentages, and generating reports are performed manually by faculty or administrative staff. This process is time-consuming and often leads to delays in preparing accurate attendance reports, especially when the number of students is large.

Manual handling of attendance data is also prone to human errors and inconsistencies. Different staff members may record or calculate attendance differently, which can result in incorrect attendance percentages and inaccurate reports. This may lead to improper identification of students with low attendance, affecting decision-making. Additionally, repetitive data entry and calculations increase workload and reduce overall productivity.

The existing system lacks proper tracking and real-time visibility, making it difficult to monitor attendance records efficiently. There is also a high dependency on human effort, which increases the chances of errors and limits the ability to manage large volumes of data effectively. Generating reports manually further adds to the complexity and delays.

As institutional requirements continue to grow, the manual approach becomes inefficient and creates scalability challenges. These limitations highlight the need for an automated system that can streamline attendance processing, improve accuracy, and ensure faster and more reliable report generation.

3.2 PROPOSED SYSTEM

To overcome the limitations of manual attendance management, the proposed system introduces an automated solution using UiPath Studio. This system is designed to extract attendance data from Excel sheets or databases, validate the data, calculate attendance percentages, and generate structured reports automatically. UiPath Studio provides a user-friendly interface with drag-and-drop features, making it easy to design and implement automation workflows without requiring advanced programming knowledge.

By integrating this automation into the existing process, institutions can significantly reduce manual effort and improve accuracy in attendance calculation and reporting. The system follows a rule-based approach, where attendance criteria such as total working days, present days, and attendance thresholds are maintained in a structured format. This allows easy updates and flexibility without modifying the core workflow. It ensures consistent calculations and faster processing of attendance data, even for large numbers of students.

The proposed system also includes features such as data validation, logging, error handling, and structured report generation, which help in maintaining reliability and monitoring system performance. Automation reduces delays, minimizes human errors, and improves overall efficiency in attendance management.

In summary, the implementation of this automated attendance extraction and report generation system using UiPath provides an efficient, scalable, and reliable solution. It enhances administrative efficiency, supports better monitoring of student attendance, and helps institutions manage attendance data more effectively in a structured and organized manner.

3.3 HARDWARE ENVIRONMENT:

- Operating System – Windows 10 (64-bit)
- Processor – Multicore processor with 2 GHz or higher
- Memory (RAM) – Minimum 8 GB
- Storage – Sufficient free space for UiPath and attendance data files
- Display – Resolution of 1920×1080
- Software – UiPath Studio (Version 2023.12.0)

3.4 SOFTWARE ENVIRONMENT:

- UiPath Studio (2023.12.0)
- UiPath Robot
- UiPath Orchestrator
- UiPath Assistant
- Configuration Management Tools
- Document Automation Tools
- Microsoft Excel

CHAPTER 4

SYSTEM DESIGN

CHAPTER – 4

SYSTEM DESIGN

4.1. FLOW CHART

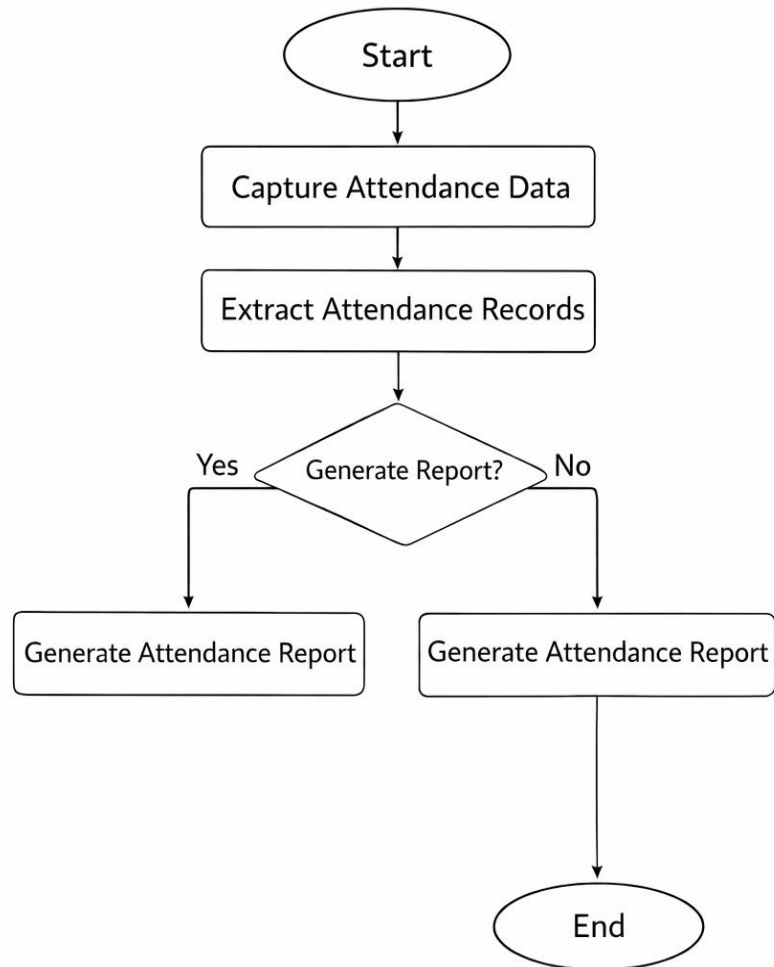


Figure 4.1 Flow Chart for Automated Attendance Extraction and Report Generation System.

4.2 ARCHITECTURE DIAGRAM

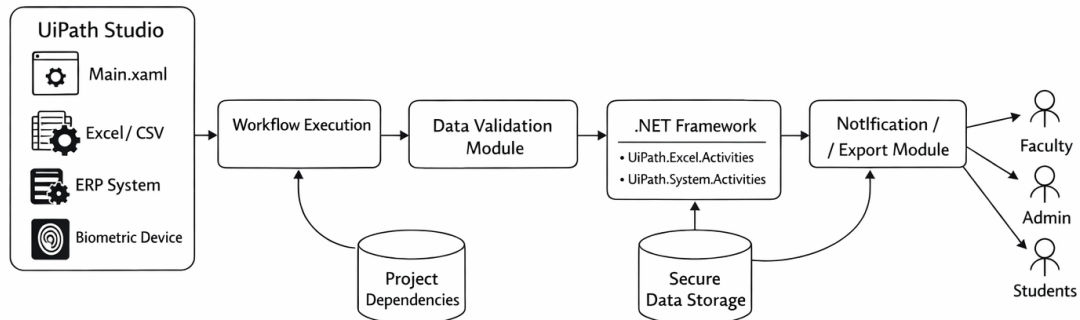


Figure 4.2 Architecture Diagram of Automated Attendance Extraction and Report Generation System.

EXPLANATION:

The above flowchart illustrates the operational workflow of the Automated Attendance Extraction and Report Generation System. It provides a clear view of how attendance data moves through different stages of the automation process, from input to final report generation.

The process begins with the collection of attendance data from Excel sheets, ERP systems, or biometric devices. At the core of the system is the automation engine, developed using UiPath/Python, which performs key operations such as data extraction, validation, and attendance percentage calculation. This ensures that attendance records are processed accurately and consistently without manual intervention.

The next stage involves report generation, where validated data is compiled into structured daily, weekly, or monthly reports. These reports are automatically exported in PDF or Excel formats for faculty and management review. Overall, the flowchart demonstrates a streamlined and efficient workflow that minimizes

human error, enhances transparency, and supports digital transformation in academic attendance management.

The above architecture diagram represents the structural design and working of the Automated Attendance Extraction and Report Generation System. It provides a detailed view of how different modules interact to achieve end-to-end automation. The process begins with attendance data sources such as Excel/CSV files, ERP systems, and biometric devices.

The **Data Collection Module** gathers raw data, which is then passed to the **Data Validation Module** for cleaning and consistency checks. The **Attendance Processing Engine** calculates attendance percentages and identifies students below the threshold. The **Report Generation Module** compiles this processed data into structured reports, while the **Notification/Export Module** distributes them to faculty, admin, and students.

In the UiPath-based implementation, the workflow is executed through Main.xaml, utilizing dependencies like UiPath.Excel.Activities and UiPath.System.Activities under the .NET framework. This ensures smooth integration, secure data handling, and reliable automation. Overall, the architecture demonstrates a modular, scalable, and efficient design that automates attendance management, reduces manual workload, and enhances institutional productivity.

4.3 DATA FLOW DIAGRAM

A Data Flow Diagram (DFD) is used to specify the functions of an information system and to illustrate how data moves from one process to another. It focuses on the flow of data rather than control flow—meaning it does not include decision rules, loops, or conditional logic. Each process in the DFD represents a transformation of data, showing how inputs are converted into outputs. Specific

operations or decision-making steps are instead represented using flowcharts. As part of structured analysis modeling tools, the DFD provides a clear and systematic way to visualize the logical flow of information within the system.

4.3.1 0 LEVEL DFD

Figure explains the overall functionality of the proposed system

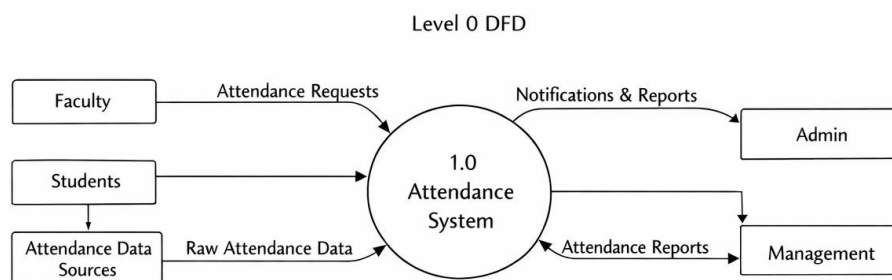


Figure 4.3.1 Data Flow Diagram at the Initial Level

4.3.2 FIRST LEVEL DFD

The Level 1 DFD shows how data moves through five modules collection, validation, processing, reporting, and notification within the Automated Attendance Extraction and Report Generation System. It captures, verifies, and processes attendance data from various sources, generates structured reports, and automatically shares them with faculty, admin, and management for efficient and transparent tracking.

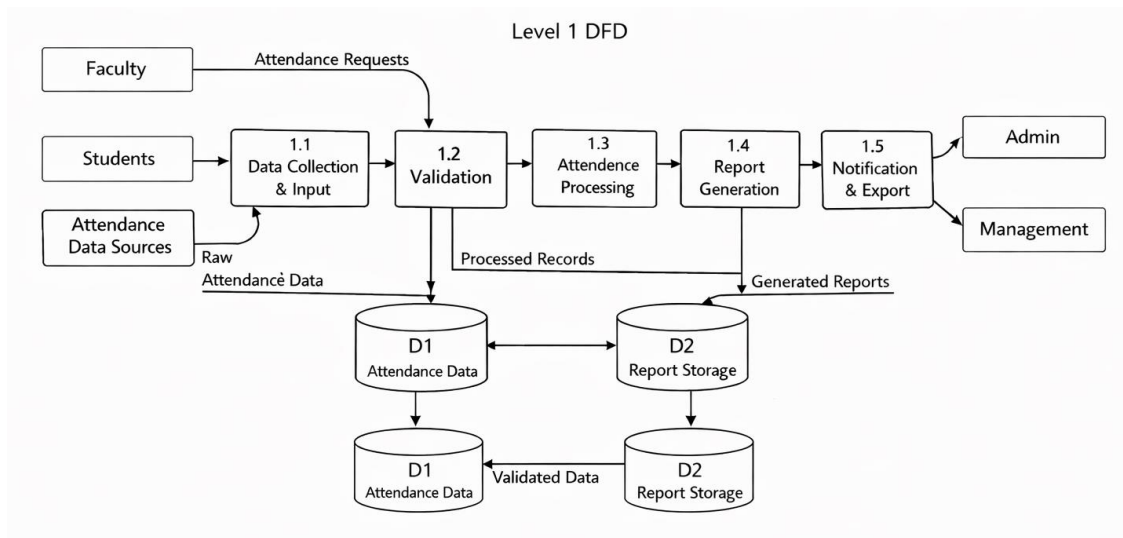


Figure 4.3.2 Data Flow Diagram Level 1

4.3.3 SECOND LEVEL DFD

The Level 2 DFD gives a concise view of the detailed internal flow within the Automated Attendance Extraction and Report Generation System. It breaks down attendance processing into calculating attendance, identifying low attendance, and generating alerts, while report generation includes creating, formatting, and saving reports. Data moves between D1 Attendance Data and D2 Report Storage, with outputs sent to Admin and Management, ensuring accurate, efficient, and transparent attendance handling.

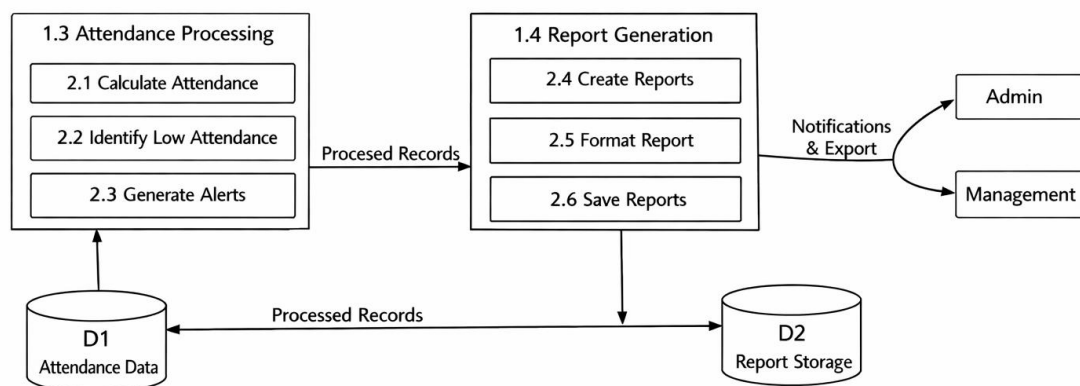


Figure 4.3.3 Data Flow Diagram Level 2

4.4 USE CASE DIAGRAM

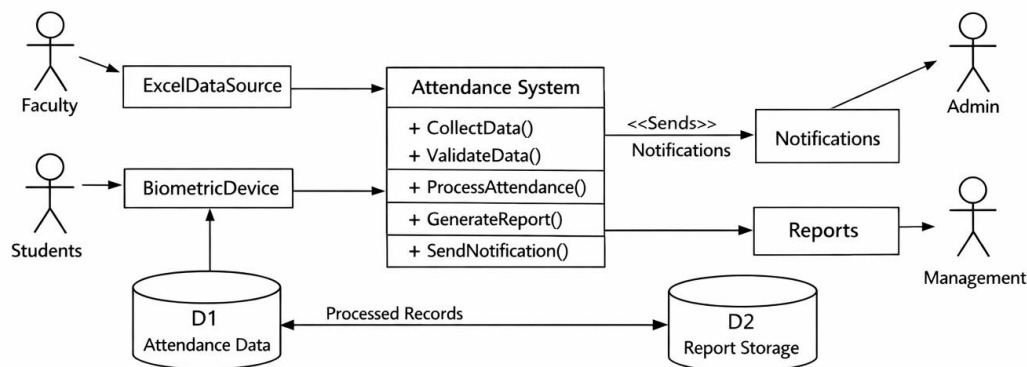


Figure 4.4 Use Case diagram of Automated Attendance Extraction and Report Generation System.

The UML diagram visually represents the structure of the Automated Attendance Extraction and Report Generation System using classes and actors. Faculty and Students interact with the system to provide attendance data, while the central Attendance System class performs operations such as CollectData(), ValidateData(), ProcessAttendance(), GenerateReport(), and SendNotification(). The ExcelDataSource and BiometricDevice classes are associated with the system for data input, and Admin and Management receive notifications and reports generated by the system. This diagram effectively captures the relationships and responsibilities of each component, making it ideal for inclusion in the System Design section of the report.

4.5 SEQUENCE DIAGRAM

The sequence diagram illustrates the chronological interaction between components of the Automated Attendance Extraction and Report Generation System.

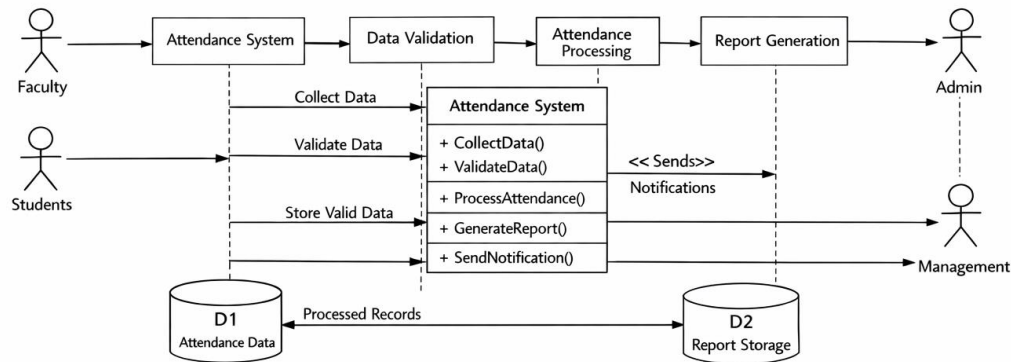


Figure 4.5 Sequence Diagram of Automated Attendance Extraction and Report Generation System.

Faculty initiates the process by submitting an attendance request, after which the Attendance System collects and validates data through the Data Validation module. The validated data is processed by the Attendance Processing module to calculate attendance and identify low attendance cases. The Report Generation module then creates, formats, and saves reports, which are finally sent to Admin and Management for review. This diagram captures the step-by-step communication flow, emphasizing automation, accuracy, and transparency in attendance management.

4.6 ACTIVITY DIAGRAM

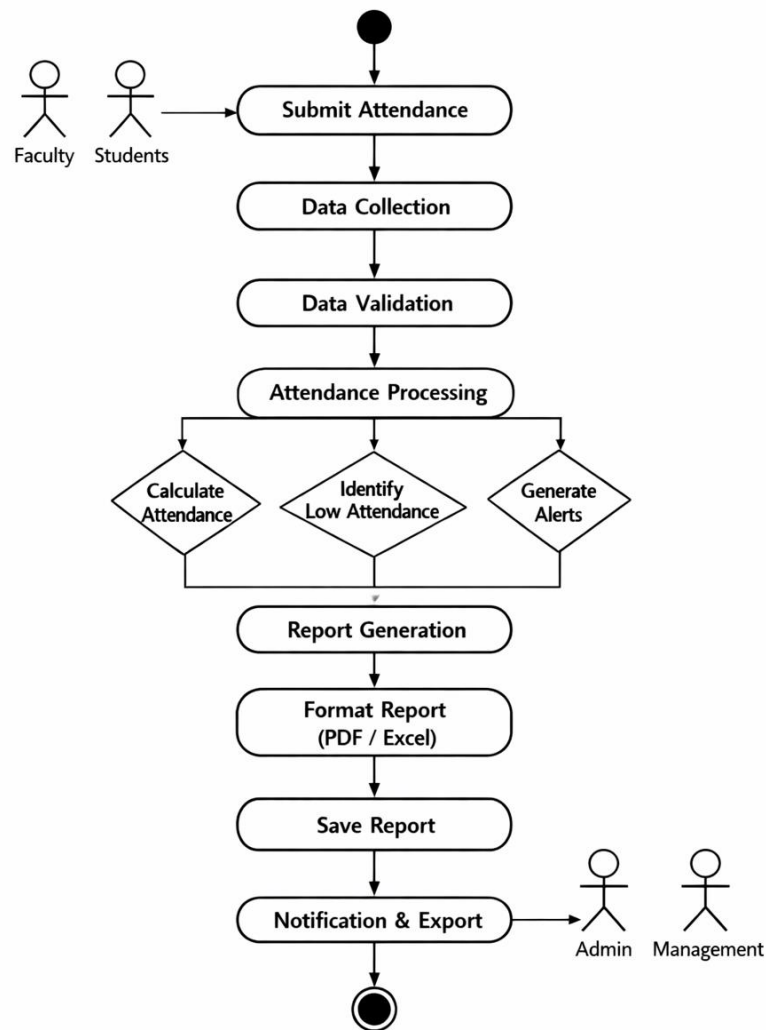


Figure4.6 Activity diagram of Automated Attendance Extraction and Report Generation System.

The activity diagram presents the step-by-step workflow of the Automated Attendance Extraction and Report Generation System. It begins with Faculty or Students submitting attendance, followed by data collection from Excel, ERP, or biometric sources. The system then validates the data, processes attendance to calculate percentages and identify low attendance cases, generates and formats reports, saves them securely, and finally sends notifications and reports to Admin and Management. This diagram highlights the sequential

execution of tasks, ensuring automation, accuracy, and transparency throughout the attendance management process.

4.7 CLASS DIAGRAM

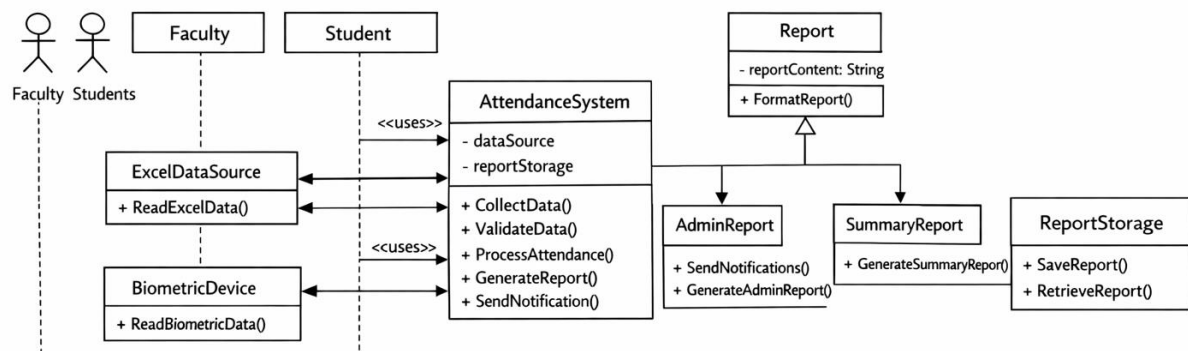


Figure4.6 Class diagram of Automated Attendance Extraction and Report Generation System.

It depicts the system's structural design through interconnected classes. The central **AttendanceSystem** class contains attributes like `dataSource` and `reportStorage`, and methods such as `CollectData()`, `ValidateData()`, `ProcessAttendance()`, `GenerateReport()`, and `SendNotification()`. Supporting classes — **ExcelDataSource** and **BiometricDevice** — provide data input through their respective methods `ReadExcelData()` and `ReadBiometricData()`. On the output side, the abstract **Report** class defines the structure for reports, with subclasses **AdminReport** and **SummaryReport** handling specific reporting tasks. The **ReportStorage** class manages saving and retrieving reports.

CHAPTER 5

SYSTEM IMPLEMENTATION

CHAPTER-5

SYSTEM IMPLEMENTATION

Implementing this project in a structured workflow involves designing and executing modules that collectively automate attendance management. The system begins with **Data Collection**, where attendance records are extracted from sources such as Excel sheets, ERP systems, or biometric devices.

This raw data is then passed to the **Data Validation** module, which checks for missing or duplicate entries to ensure accuracy and reliability. Once validated, the workflow transitions to **Attendance Processing**, where the system calculates attendance percentages, identifies students with low attendance, and generates alerts for intervention.

The **Report Generation** module compiles this processed data into structured reports, formats them into PDF or Excel files, and stores them securely.

Finally, the **Notification & Export** module distributes detailed reports to Admin and summarized reports to Management, ensuring transparency and supporting decision-making. Throughout the workflow, error handling and logging mechanisms are integrated to maintain smooth execution and traceability.

This end-to-end automation covers the complete lifecycle of attendance management — from data extraction and validation to processing, reporting, and notification. It eliminates manual effort, ensures accuracy, and enhances operational efficiency. The system is scalable, capable of handling large volumes of attendance data, and provides a reliable, secure, and high-performance solution for modern institutional needs.

5.1 WORKFLOW

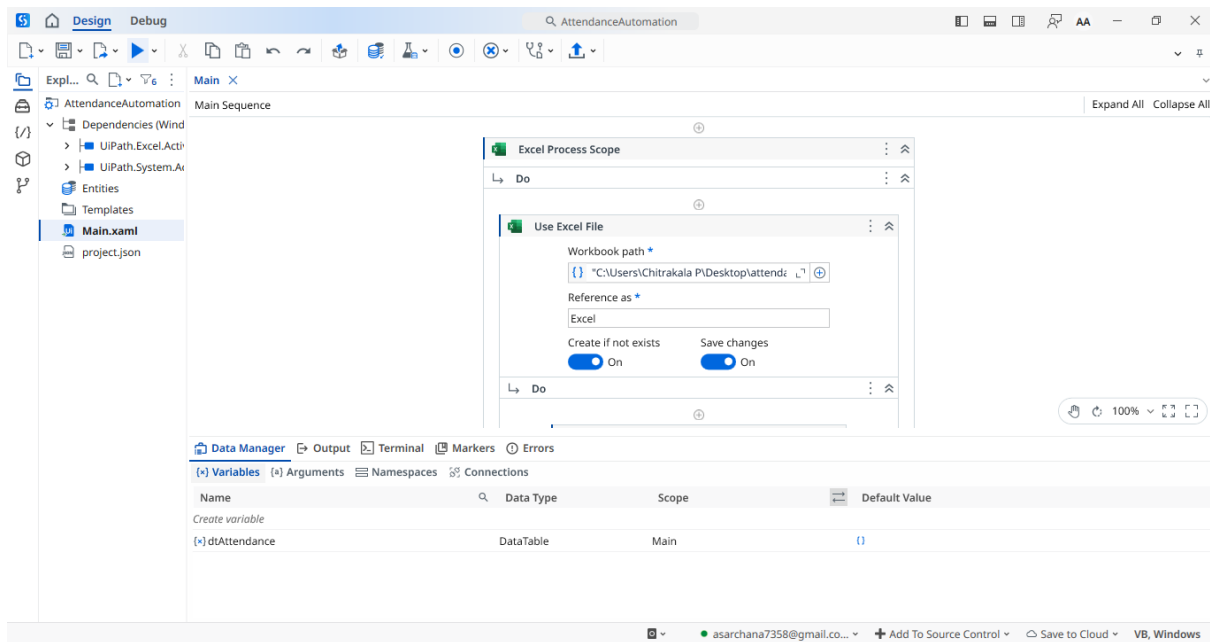


Fig 5.1.1: Main Workflow Overview

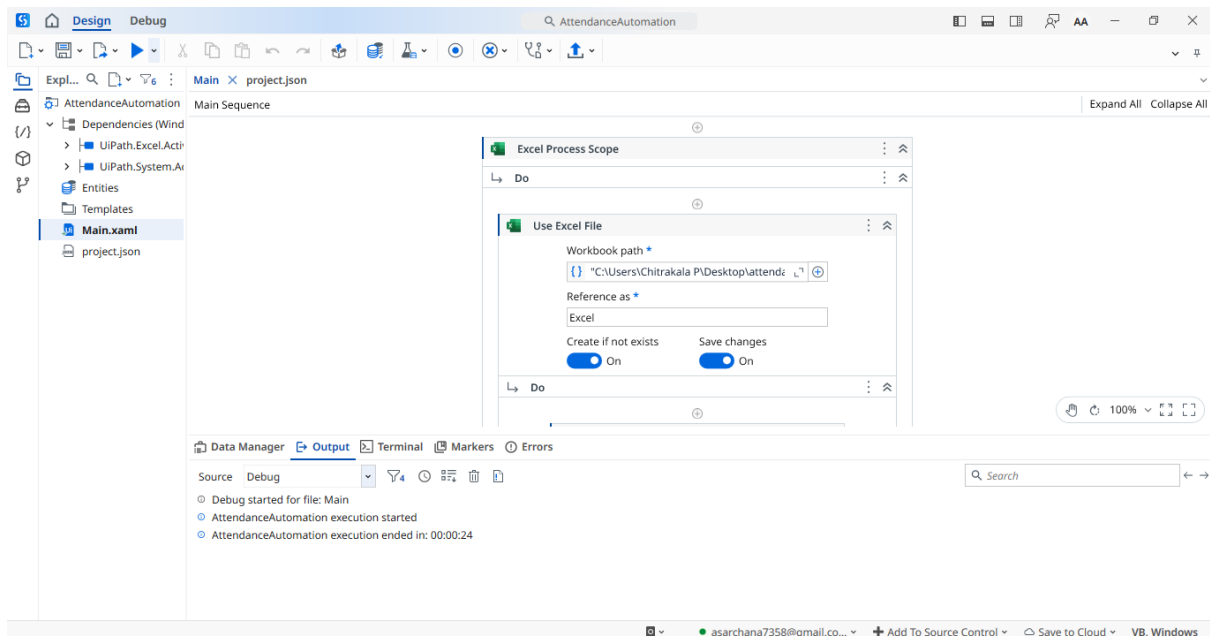


Fig 5.1.2 Write Output Module

CHAPTER 6

SYSTEM TESTING

6.1 TEST CASES AND REPORT

1. Check data extraction from Excel/database

The system should correctly extract attendance data from Excel sheets or databases. This ensures accurate input for further processing.

2. Verify data validation process

The system should identify missing, duplicate, or incorrect data entries. This ensures only valid data is processed.

3. Validate attendance calculation

The system should correctly calculate attendance percentage based on total days and present days. This ensures accuracy in results.

4. Check low attendance identification

The system should correctly identify students below the attendance threshold. This helps in monitoring and decision-making.

5. Ensure report generation (Excel/PDF)

The system should generate structured attendance reports in Excel or PDF format. This ensures proper documentation.

6. Test report accuracy

Generated reports should match the processed attendance data. This ensures correctness and reliability.

7. Verify notification/export module

Reports should be successfully sent to admin/management. This ensures proper communication.

8. Check system response time

The system should process data and generate reports within a few seconds. This ensures efficiency.

9. Validate error handling

The system should handle errors like missing files or incorrect formats gracefully. This ensures robustness.

10. Check system scalability

The system should handle large datasets without crashing. This ensures performance under heavy load

Table 6.1 Testcases and Report

Test Case ID	Test Scenario	Expected Result	Actual Result	Pass/Fail
TC1	Check data extraction from Excel/database	System extracts attendance data correctly	Data extracted successfully	Pass
TC2	Verify data validation process	Invalid or missing data is detected	System identifies errors correctly	Pass
TC3	Validate attendance calculation	Correct percentage is calculated	Accurate calculation performed	Pass
TC4	Check low attendance identification	Students below threshold are identified	Low attendance students listed correctly	Pass
TC5	Ensure report generation	Reports generated in Excel/PDF format	Reports generated successfully	Pass
TC6	Test report accuracy	Report data matches processed data	Data verified and correct	Pass

TC7	Verify notification/export module	Reports sent to admin/management	Reports delivered successfully	Pass
TC8	Check system response time	System responds within seconds	Fast response observed	Pass
TC9	Validate error handling	System handles errors without crashing	Errors managed properly	Pass
TC10	Check system scalability	System handles large data efficiently	System performs without failure	Pass

CHAPTER 7

CONCLUSION

CHAPTER-7

CONCLUSION

In conclusion, the Automated Attendance Extraction and Report Generation System stands as a comprehensive demonstration of how academic institutions can leverage structured automation to overcome inefficiencies in traditional attendance management. By carefully integrating modules for data collection, validation, processing, reporting, notification, and storage, the system ensures that every stage of the workflow is executed with precision, reliability, and transparency. The layered design, supported by DFDs, UML diagrams, sequence diagrams, and activity charts, reflects not only the robustness of the architecture but also its adaptability to diverse institutional requirements. This project eliminates manual errors, reduces administrative workload, and enhances decision-making by delivering timely insights to management and detailed reports to administrators. Its automation capabilities minimize human intervention while supporting scalability, making it capable of handling large datasets and multiple input sources with ease. Ultimately, this project is not just a technical solution but a strategic step toward modernizing academic governance, fostering accountability, and ensuring that institutions are equipped with reliable tools to meet the evolving demands of education management.

FUTURE SCOPE

Future improvements can include integration of biometric or face recognition systems for real-time attendance collection without manual input. Cloud-based implementation can enable remote access and better scalability across institutions. Machine learning techniques can be used to analyze attendance patterns and identify students at risk. Adding mobile application support will allow faculty to manage attendance and view reports easily. Integration with ERP

and LMS systems can ensure smooth data flow, while automated email or SMS notifications can alert students about low attendance. These enhancements will make the system more efficient, intelligent, and user-friendly.

APPENDICES

A.1 Code snippet

```
{  
  "name": "AttendanceAutomation",  
  "projectId": "3e2bde2a-120b-46a0-8d7f-506f7af27c55",  
  "description": "Blank Process",  
  "main": "Main.xaml",  
  "dependencies": {  
    "UiPath.Excel.Activities": "[3.4.1]",  
    "UiPath.System.Activities": "[26.2.4]"  
  },  
  "webServices": [],  
  "entitiesStores": [],  
  "schemaVersion": "4.0",  
  "studioVersion": "26.0.190.0",  
  "projectVersion": "1.0.0",  
  "runtimeOptions": {  
    "autoDispose": false,  
    "netFrameworkLazyLoading": false,
```

```
"isPausable": true,

"isAttended": false,

"requiresUserInteraction": true,

"supportsPersistence": false,

"workflowSerialization": "NewtonsoftJson",

"excludedLoggedData": [

  "Private:*",

  "*password*"

],

"executionType": "Workflow",

"readyForPiP": false,

"startsInPiP": false,

"mustRestoreAllDependencies": true,

"pipType": "ChildSession"

},

"designOptions": {

  "projectProfile": "Developement",

  "outputType": "Process",

  "libraryOptions": {

    "privateWorkflows": []

  },

  "processOptions": {
```

```

    "ignoredFiles": [],
  },
  "fileInfoCollection": [],
  "saveToCloud": false
},
"expressionLanguage": "VisualBasic",
"entryPoints": [
  {
    "filePath": "Main.xaml",
    "uniqueId": "90350059-f613-464d-ba03-bd6faf59c83b",
    "input": [],
    "output": []
  }
],
"isTemplate": false,
"templateProjectData": {},
"publishData": {},
"targetFramework": "Windows"
}

```

FRONTEND CODE (HTML + CSS)

```
<!DOCTYPE html>
```

```
<html>
```

```

<head>

<meta charset="UTF-8">

<title>Attendance Report</title>

<style>

/* — Base — */

body {

    margin: 0;

    font-family: Arial, sans-serif;

    background: #f0f2f5;

}


/* — Header — */

header {

    background: #1a237e;

    color: #fff;

    padding: 24px 20px;

    text-align: center;

}

header h1 { margin: 0 0 4px; font-size: 22px; }

header p { margin: 0; opacity: .75; font-size: 13px; }


/* — Wrapper — */

```



```
.wrap {  
  
  max-width: 960px;  
  
  margin: 24px auto;  
  
  padding: 0 20px;  
  
}  
  
/* — Summary Cards — */  
  
.cards {  
  
  display: flex;  
  
  gap: 14px;  
  
  margin-bottom: 22px;  
  
}  
  
.card {  
  
  flex: 1;  
  
  background: #fff;  
  
  border-radius: 10px;  
  
  padding: 18px;  
  
  text-align: center;  
  
  box-shadow: 0 2px 6px rgba(0,0,0,.1);  
  
}  
  
.lbl {  
  
  font-size: 11px;
```

```

color: #888;

text-transform: uppercase;

letter-spacing: .5px;

margin-bottom: 6px;
}

.num { font-size: 30px; font-weight: 700; }

/* — Table — */

table {

width: 100%;

background: #fff;

border-radius: 10px;

overflow: hidden;

box-shadow: 0 2px 6px rgba(0,0,0,.1);

border-collapse: collapse;

}

th {

background: #1a237e;

color: #fff;

padding: 12px 14px;

text-align: left;

font-size: 11px;

```

```

    text-transform: uppercase;

}

td {

    padding: 11px 14px;

    border-bottom: 1px solid #f0f2f5;

    font-size: 13px;

}


/* — Row colours — */

.rp { background: #f9fbe7; } /* Present row — light green */

.ra { background: #fff8f8; } /* Absent row — light red */


/* — Status badges — */

.bp {

    display: inline-block;

    padding: 2px 10px;

    border-radius: 20px;

    background: #c8e6c9;

    color: #2e7d32;

    font-size: 11px;

    font-weight: 700;

}

```

```

.ba {
    display: inline-block;
    padding: 2px 10px;
    border-radius: 20px;
    background: #ffcdd2;
    color: #c62828;
    font-size: 11px;
    font-weight: 700;
}
</style>
</head>
<body>

<!-- Header -->
<header>
    <h1>Attendance Report</h1>
    <p>Automated Attendance Management System</p>
</header>

<div class="wrap">

<!-- Summary Cards (values injected dynamically by UiPath) -->

```

```

<div class="cards">

  <div class="card">

    <div class="lbl">Total Students</div>

    <div class="num" style="color:#1a237e"><!-- dtAttendance.Rows.Count -
-></div>

  </div>

  <div class="card">

    <div class="lbl">Present</div>

    <div class="num" style="color:#2e7d32"><!-- Present count --></div>

  </div>

  <div class="card">

    <div class="lbl">Absent</div>

    <div class="num" style="color:#c62828"><!-- Absent count --></div>

  </div>

</div>

<!-- Data Table -->

<table>

  <thead>

    <tr>

      <th>#</th>

      <th>Name</th>


```

```

    <th>Present Days</th>

    <th>Total Days</th>

    <th>Percentage</th>

    <th>Status</th>

</tr>

</thead>

<tbody>

    <!-- Rows injected dynamically via LINQ by UiPath -->

    <!-- Example of a generated row: -->

    <tr class="rp">

        <td>1</td>

        <td><b>Student Name</b></td>

        <td>22</td>

        <td>25</td>

        <td><b>88%</b></td>

        <td><span class="bp">Present</span></td>

    </tr>

    <tr class="ra">

        <td>2</td>

        <td><b>Student Name</b></td>

        <td>15</td>

        <td>25</td>

```

```
<td><b>60%</b></td>
```

```
<td><span class="ba">Absent</span></td>
```

```
</tr>
```

```
</tbody>
```

```
</table>
```

```
</div>
```

```
</body>
```

```
</html>
```

output

The screenshot shows a web browser window with three tabs, all titled 'Attendance Report'. The address bar shows the file path: C:/Users/Chitrakala%20P/Documents/UiPath/AttendanceAutomation/AttendanceReport.html. The application has a dark blue header with the title 'Attendance Report' and subtitle 'Automated Attendance Management System'. Below the header, there are three summary cards: 'TOTAL STUDENTS' with the value 40, 'PRESENT' with the value 18, and 'ABSENT' with the value 22. Below these cards is a table with 10 rows of student data. The table has columns for ID, Name, Present Days, Total Days, Percentage, and Status. The status is indicated by a green 'Present' button or a red 'Absent' button.

#	NAME	PRESENT DAYS	TOTAL DAYS	PERCENTAGE	STATUS
1	Ravi	25	30	83.33%	Present
2	Anu	18	30	60%	Absent
3	Kiran	28	30	93.33%	Present
4	Meena	20	30	66.67%	Absent
5	Arjun	10	30	33.33%	Absent
6	Priya	27	30	90%	Present
7	Suresh	15	30	50%	Absent
8	Divya	29	30	96.67%	Present
9	Sanjay	22	30	73.33%	Absent
10	Lakshmi	30	30	100%	Present

Figure 7.1.1 Processed Output

This screenshot shows a continuation of the attendance report, displaying rows 10 through 25 of the student list. The layout is consistent with the previous screenshot, showing a table with columns for ID, Name, Present Days, Total Days, Percentage, and Status. The status is indicated by a green 'Present' button or a red 'Absent' button.

10	Lakshmi	30	30	100%	Present
11	Rohit	16	30	53.33%	Absent
12	Sneha	24	30	80%	Present
13	Manoj	19	30	63.33%	Absent
14	Kavya	21	30	70%	Absent
15	Ajay	26	30	86.67%	Present
16	Neha	14	30	46.67%	Absent
17	Varun	23	30	76.67%	Present
18	Ramesh	12	30	40%	Absent
19	Shalini	20	30	66.67%	Absent
20	Vivek	17	30	56.67%	Absent
21	Pooja	28	30	93.33%	Present
22	Harish	13	30	43.33%	Absent
23	Aarti	25	30	83.33%	Present
24	Nikhil	22	30	73.33%	Absent
25	Rekha	18	30	60%	Absent

Figure 7.1.2 Processed Output

25	Rekha	18	30	60%	Absent
26	Ashok	27	30	90%	Present
27	Jyoti	29	30	96.67%	Present
28	Deepak	11	30	36.67%	Absent
29	Ritu	19	30	63.33%	Absent
30	Mohan	15	30	50%	Absent
31	Geeta	24	30	80%	Present
32	Tarun	20	30	66.67%	Absent
33	Anjali	26	30	86.67%	Present
34	Karthik	23	30	76.67%	Present
35	Swati	30	30	100%	Present
36	Parth	16	30	53.33%	Absent
37	Radha	21	30	70%	Absent
38	Sameer	28	30	93.33%	Present
39	Bhavna	13	30	43.33%	Absent
40	Omkar	25	30	83.33%	Present

Figure 7.1.3 Processed Output

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